Computer Vision Based Road Safety Monitoring System

Project Report: Image Processing and Computer Vision (IPCV)

Team 21

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Abstract

Road accidents are a serious issue in many cities, often caused by people ignoring traffic rules like wearing helmets or riding with three people on a bike. Manually checking traffic cameras is too slow and prone to errors. To solve this, we built a Computer Vision-Based Road Safety Monitoring System that automatically spots traffic violations and road hazards from images. The system features four fully completed modules: helmet detection, triple-riding detection, pothole detection, and automatic number plate recognition (ANPR). We built the detection models using YOLOv8, OpenCV, and PyTorch. Everything is connected through a FastAPI backend to a React dashboard, where users can upload an image and instantly see violations highlighted with bounding boxes.

1. Introduction

Two-wheelers are very common on Indian roads, and a large number of accidents involve riders without helmets. Triple riding and poor road conditions like potholes also cause many accidents. Right now, authorities mostly rely on manual CCTV checks, which is inefficient.

Objectives:

- Detect riders with and without helmets using deep learning.
- Detect triple riding on two-wheelers.
- Identify potholes on the road.
- Read vehicle number plates using ANPR to identify the violator.
- Build a simple web dashboard to easily view the results.

2. Project Modules & Team

The work was split among our four team members. All modules are now completely developed and integrated into the final dashboard.

Module	Team Member	Implementation Details
Helmet Violation Detection	Vishal Patel	Used YOLOv8n trained on a 2-class helmet dataset.
Triple-Riding Detection	Sojwal Vilhekar	Detects bikes and passengers, then counts the riders on each bike.
Pothole Detection	Virajsingh Rajput	Used YOLO trained on a pothole dataset.
ANPR	Shibchandan Mistry	Detects the number plate with YOLO and reads it using EasyOCR.

3. Technologies Used

- **Python & OpenCV:** Main language and image processing.
- **YOLOv8 (Ultralytics) & PyTorch:** Object detection model and deep learning framework.
- **NumPy:** Handling image arrays.
- **EasyOCR:** Reading text from number plates.
- **FastAPI and Uvicorn:** Backend API and server.
- **React with Vite:** Frontend dashboard.

4. System Architecture

Our system is broken into three layers so everything can run smoothly:

1. **Frontend (React):** The user uploads a traffic image. The dashboard shows stats and draws red boxes for violations and green boxes for safe riders.
2. **Backend (FastAPI):** It receives the image from the browser, passes it to the AI script, counts the violations, and sends the JSON results back.
3. **Computer Vision Layer (YOLOv8 + PyTorch):** This contains the trained models that actually find the riders, potholes, and plates.

5. Dataset and Model Training

For helmet detection, we used a dataset of 1,376 images split into train, valid, and test sets. We also used specific datasets for potholes and a 3-class dataset (Bike, Helmet, Passenger) for triple-riding. We used YOLOv8 Nano because it is lightweight and runs easily on a standard laptop. By using transfer learning on weights already trained on the COCO dataset, we fine-tuned the model on our custom data for 50 epochs, saving a lot of time.

6. Implementation

- **Detection Script:** The script loads the trained model weights and runs predictions. It returns the box coordinates, the class name, and confidence score. If it detects a violation, it flags it.
- **Backend API:** Built with FastAPI. It handles the image uploads, runs them through the detection scripts, and securely deletes the temporary images afterward.
- **Frontend Dashboard:** A dark-themed web app where users can drag and drop images. It features a detection table, a confidence threshold slider, and allows users to download the annotated result.
- **Integrated Modules:** The system now seamlessly handles triple-riding (checking if three or more riders overlap a single bike), identifies potholes, and uses EasyOCR to crop and read vehicle number plates of violators all in one go.

7. Results and Limitations

The system quickly processes images and correctly updates the dashboard statistics in real-time. The confidence slider helps users adjust how sensitive the detections are based on the camera view.

Limitations:

- Since our datasets are relatively small, the models might struggle in poor lighting, rain, or difficult camera angles.
- Very distant or blurry riders can sometimes be missed.
- The violations are currently not stored in a database for long-term records.

8. Conclusion

We successfully developed a full Computer Vision-Based Road Safety Monitoring System. By utilizing YOLOv8 and transfer learning, we integrated helmet detection, triple riding, potholes, and ANPR into a single system connected via FastAPI to a React interface. This project taught us how to build a complete pipeline from model training to frontend deployment. We believe this system can be a great tool to help authorities monitor roads more efficiently.

References

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2. Ultralytics, YOLOv8 Documentation
3. JaidedAI, EasyOCR
4. Project repository: https://github.com/vishalpatel2023/TEAM_21